SALAH ELBAKRI

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EDUCATION

University of Washington, Seattle *09/2023 – 03/2025*
M.S. in Data Science

University of Massachusetts, Amherst *09/2014 – 05/2017*
B.S. in Electrical Engineering

CORE SKILLS

- **Languages:** C++, C, Python, SQL, Bash, Node.js
- **Embedded Systems & Hardware:** Microcontrollers, Arduino, RTOS, Firmware, Device Drivers, Bare-Metal, ARM, DSP
- **Architecture & Cloud:** Distributed Systems, Microservices, AWS, Docker, Kubernetes, High-Concurrency
- **DevOps & Tooling:** Terraform, CI/CD pipelines, Linux/UNIX shell, Git, pnpm, Moon

EXPERIENCE

ISO New England *05/2019 – 03/2022*
Power Systems Engineer

- Engineered analytics pipelines to process 5TB+ of time-series data, enabling real-time system monitoring and operational excellence for critical grid infrastructure.
- Provisioned fault-tolerant infrastructure using Terraform and UNIX/Linux shell scripting, ensuring secure and reproducible deployments in a distributed computing environment.
- Automated contingency checks and maintained resilient systems, reducing manual data verification efforts by **85%** and minimizing production issues.
- Collaborated cross-functionally to orchestrate distributed workflows using GitHub CI/CD, enabling **30%** faster release cycles in an agile development environment.

Bonneville Power Administration *09/2024 – 03/2025*
Business/Financial Analyst

- Engineered data models to track system transformations, increasing data observability by **80%** and streamlining operational forecasting.
- Automated ETL pipelines for multi-source data integration, rigorously improving data quality control (QC) and reducing manual processing time by **40%**.

Regeneron Pharmaceuticals *06/2017 – 04/2019*
Engineering Consultant, Advanced Analytics

- Built anomaly detection models integrating equipment sensor data to perform rigorous quality control on manufacturing lines, reducing downtime by **25%**.
- Collaborated directly with bioprocessing engineers to translate raw machine data into actionable, automated insights for continuous operational improvement.

University of Massachusetts, Amherst *09/2016 – 05/2017*
Lab Teaching Assistant, Circuits & Semiconductors

- Mentored students in circuit design, microcontroller integration, and semiconductor principles, reinforcing embedded systems intuition and hardware troubleshooting skills.
- Evaluated practical hardware and firmware implementations, ensuring adherence to rigorous engineering standards.

PROJECTS

Brain-Controlled Prosthetic System | *C++, Arduino, Embedded Systems, DSP*

- Developed embedded firmware in C++ for an Arduino microcontroller to acquire, filter, and process EEG wavelets, maximizing SNR for real-time control of a mechanical arm.
- Engineered a low-latency, bare-metal embedded system translating neurological signals into mechanical actuation, achieving **85%** accuracy at sub-100ms response times.
- Designed the hardware-software interface, directly integrating sensor inputs with microcontroller logic to drive the mechanical arm's multi-axis movements seamlessly.

Flipkart Customer Support AI Agent | *Python, Azure OpenAI, RAG*

- Built a high-concurrency, production-grade multi-agent LLM solution to automate problem-solving and workflow resolution while following standard operating procedures.
- Integrated Retrieval-Augmented Generation (RAG) to maintain data context, delivering a **35%** improvement in automated resolution rates.

Shopify E-Commerce Database Architecture | *SQL, Python, ETL, Data Modeling*

- Designed a normalized relational database schema integrating five distinct e-commerce domains: transaction ledgers, customer profiles, product catalogs, 3PL inventory logistics, and gift card balances.
- Engineered scalable ETL pipelines to process over 150 business attributes per order, optimizing data structures for high-volume analytics and reducing reporting latency by **30%**.

Brain Tumor Detection Pipeline | *Python, PyTorch, UNIX, Spatial Data*

- Managed an end-to-end workflow in a UNIX environment from massive data ingestion to model training, achieving **15%** higher accuracy.